

DigitalOcean Application Stack

Customer Recommendations Report

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1. What Was Built

A working reference implementation deployed on DigitalOcean, connecting a web application to two managed databases.

Resource	Component / Tier	Where
Web application	my-app (basic-xxs)	nyc
Relational DB	my-app-pg (PostgreSQL 18)	nyc1
Cache / store	my-app-valkey (Valkey 8)	nyc1

Live URL: <https://my-app-2x7kf.ondigitalocean.app>

Source: <https://github.com/srihariprabhakar/my-app>

Stack: Node.js (Express), PostgreSQL, Valkey (Redis-compatible)

Verified end-to-end:

- GET /health -> {"checks":{"postgres":"ok","valkey":"ok"}}
- POST /visit -> writes a PostgreSQL record and increments a Valkey counter
- App Platform reports the web component as HEALTHY

2. Cost Estimate

Approximate monthly list prices for this exact configuration:

Resource	Tier	USD / mo
App Platform service	basic-xxs (1 instance)	~\$5
Managed PostgreSQL	db-s-1vcpu-1gb	~\$15
Managed Valkey	db-s-1vcpu-1gb	~\$15
Total		~\$35

Build costs: negligible (each deploy builds in ~11-13 billable seconds, a fraction of a cent). Only the running resources generate steady monthly billing.

Note: figures are approximate U.S. list prices; confirm engine/version pricing in the billing dashboard.

3. Scaling Recommendations

- Vertical first, then horizontal: increase instance_size_slug before adding replicas.
- Scale the app out (instance_count 1 -> 2+) when CPU-bound; App Platform load-balances automatically.
- Scale the database up (e.g. db-s-2vcpu-4gb or a gd- size) before read replicas / HA.
- Add connection pooling (PgBouncer or pg pool config) as concurrency grows.
- Enable autoscaling for variable traffic; downscale when idle to control cost.

4. Reliability Assessment

This is a solid, reliable foundation for small-to-medium production workloads.

Strengths:

- Managed DBs provide automated backups, patching, snapshots, and role management.
- App Platform provides zero-downtime deploys, automatic TLS, health checks, and crash restarts.
- All resources share one region (nyc / nyc1), keeping app-to-DB latency low.

Production hardening considerations:

- Databases are single-node; add a standby / HA tier for mission-critical data.
- basic-xxs and db-s-1vcpu-1gb are dev sizes and will throttle under real load.
- Add alerting (CPU/memory thresholds) beyond DEPLOYMENT_FAILED.

5. Implementation Details

Service wiring:

Database connection info is passed via environment variables. This deployment sets DATABASE_URL (PostgreSQL) and REDIS_URL (Valkey) as encrypted SECRET env vars.

Key technical note:

DigitalOcean managed DB URIs carry sslmode=require, which node-postgres treats as verify-full and rejects on the CA chain. The app strips that parameter and supplies ssl: { rejectUnauthorized: false }.